Power Budget - Zander Boward

Team Number:	**202**
Project Name:	**Weather Gadget**
Team Member Names:	**Sidra, Brendan, Branden**
Version:	**1**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements

All Major Components	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Curiosity Nano target (PROVISIONAL)	PIC18F57Q43 Curiosity Nano	3.3 V design	1	50	50.000	mA
	LCD module incl. backlight	NHD-0420H1Z-FSW-GBW-33V3	3.0-3.6 V logic	1	17.5	17.500	mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design. Add additional power rails or change the power rail voltages if needed.

Unused power rail	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
c1. Not applicable					*Not applicable - no devices on this rail*		

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	3.3 V downstream load	From 3.3 V rail subtotal	5 V input path	1	67.5	67.500	mA

	LD1117 ground current allowance	LD1117V33	5 V	1	10	10.000	mA
						0.000	
						0.000	
						0.000	
					Subtotal	77.500	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	96.875	mA
c2. Regulator or Source Choice	5 V regulator	L7805CV	9 V nominal input	1	1500	1500.000	mA
					Total Remaining Current Available on +5V Rail	1403.125	mA

Unused power rail	**Component Name**	**Part Number**	**Supply Voltage Range**	**#**	**Absolute Maximum Current (mA)**	**Total Current (mA)**	**Unit**
c3. Not applicable					***Not applicable - no devices on this rail***		

+3.3V Power Rail	**Component Name**	**Part Number**	**Supply Voltage Range**	**#**	**Absolute Maximum Current (mA)**	**Total Current (mA)**	**Unit**
	Curiosity Nano target (PROVISIONAL)	PIC18F57Q43 Curiosity Nano	3.3 V design	1	50	50.000	mA
	LCD module incl. backlight	NHD-0420H1Z-FSW-GBW-33V3	3.0-3.6 V logic	1	17.5	17.500	mA
						0.000	
						0.000	
					Subtotal	67.500	mA
					Safety Margin	25%	

						Total Current Required on +3.3V Rail	84.375	mA
c4. Regulator or Source Choice	3.3 V regulator	LD1117V33		5 V nominal input	1	800	800.000	mA
						Total Remaining Current Available on 3.3V Rail	715.625	mA

C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available on each rail above is not negative.

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If you need multiple power sources, list each separately below and indicate which regulators will be connected to each supply. Confirm that the Total Remaining Current Available on each power source below is not negative.

External Power Source 1	**Component Name**	**Part Number**	**Supply Voltage Range**	**Output Voltage**	**Absolute Maximum Current (mA)**	**Total Current (mA)**	**Unit**
Power Source 1 Selection	9 V wall adapter	Adafruit 63	100-240 VAC	9 VDC	1000	1000.000	mA
Power Rails Connected to External Power Source 1	5 V stage demand +25%	L7805CV	9 V input path	1	96.875	96.875	mA
	L7805 ground current +25%	L7805CV	9 V	1	12.5	12.500	mA
						0.000	
					Total Remaining Current Available on External Power Source 1	890.625	mA

External Power Source 2	**Component Name**	**Part Number**	**Supply Voltage Range**	**Output Voltage**	**Absolute Maximum Current (mA)**	**Total Current (mA)**	**Unit**
Power Source 2 Selection	Not applicable - only one external source						
Power Rails Connected to External Power Source 2							

E. Calculate Battery Life (if applicable). For each battery, also check the worst-case lifetime of the battery by indicating the capacity in mAh.

Component Name	Part Number	Supply Voltage Range	Capacity (mAh)	Required By Regulators	
Not applicable - wall powered					

Notes

F9 = 50 mA is a PROVISIONAL Nano target-side allowance, including support loading; not a datasheet maximum. Clock/peripherals not finalized.

LCD F10 = 2.5 + 15 = 17.5 mA (datasheet p.5, 25 C). Both regulator ground-current inputs use 10 mA planning allowances; not full-temperature bounds.

Sources: LCD - https://newhavendisplay.com/content/specs/NHD-0420H1Z-FSW-GBW-33V3.pdf (p.5).

Regulators: https://www.st.com/resource/en/datasheet/ld1117.pdf ; https://www.st.com/resource/en/datasheet/l78.pdf . Ratings require adequate cooling.

MCU: PIC18F27/47/57Q43 datasheet, Table 47-2: 8.2 mA max at 64 MHz HFINTOSC, 3.0 V; not a bound for this 3.3 V board.